AI ASSIGNMENT WEEK 7

Q1: What is algorithmic bias, and how does it show up in real AI systems?

Algorithmic bias happens when an AI system makes unfair decisions because of the data it was trained on or how it was built. These biases often reflect real-world inequalities, so even if a model seems “neutral,” it can still make choices that aren’t fair.

Example 1: A hiring tool trained on past job applications might learn patterns that favor men over women, especially if most of the people previously hired were male. As a result, it could end up ranking female candidates lower without realizing it’s doing so.

Example 2: A credit scoring model might unintentionally give lower scores to people from certain neighbourhoods. If those areas are associated with lower income or historically underserved communities, the model could reinforce existing financial gaps.

Q2: What's the difference between transparency and explainability in AI, and why do they matter?

Transparency is about how open and clear an AI system is. It includes things like where its data came from, how the model was built, and what goals it's trying to achieve. It's like looking under the hood of the system to understand how it works.

Explainability focuses on helping people understand the decisions the AI makes. For example, if an AI denies someone a loan, explainability means breaking down the decision in plain terms so it makes sense to the person affected.

Why they matter: Without transparency, it's hard to know whether an AI is built fairly. Without explainability, people can’t challenge or learn from its decisions. Together, these qualities help build trust, especially in sensitive areas like healthcare, justice, and recruitment.

Q3: How does the GDPR (General Data Protection Regulation) impact AI development in the EU?

GDPR sets strict rules on how personal data should be collected, stored, and used. For AI developers, this means they need to be extra careful when working with datasets that include any identifiable information about people.

Here’s how it affects AI:

- **Data Collection**: Developers must get clear consent before using personal data. You can’t just scrape user info from random sources and use it to train models.
- **Right to Explanation**: If an AI system makes an automated decision that affects someone, like denying a loan or job offer, the person has the right to ask for an explanation. This pushes developers to build models that are not only accurate but also understandable.
- **Data Minimization**: AI systems should only use data that’s absolutely necessary. No hoarding massive datasets just in case, every data point must serve a clear purpose.

- **Privacy by Design**: GDPR encourages developers to build privacy protections into their systems from the ground up, rather than adding them on at the end.

Q4: Match each ethical principle to its definition.

Principles:

- A) Justice
- B) Non-maleficence
- C) Autonomy
- D) Sustainability

Definitions and Matches:

- Ensuring AI does not harm individuals or society → **B) Non-maleficence**
- Respecting users' right to control their data and decisions → **C) Autonomy**
- Designing AI to be environmentally friendly → **D) Sustainability**
- Fair distribution of AI benefits and risks → **A) Justice**

PART 2

Case 1: Biased Hiring Tool

Scenario: Amazon's AI recruiting tool penalized female candidates.

Q5: Identify the source of bias.

The bias came mainly from the training data. The tool was trained on past hiring patterns that favored male applicants. Because the model learned from this data, it picked up gendered patterns, like penalizing resumes with words linked to women's experiences or education. It wasn't intentionally built to discriminate, but it ended up reflecting past inequality.

Q6: Propose three fixes to make the tool fairer.

1. **Rebalance the training data**: Include a diverse set of resumes with equal representation across gender and other demographic lines.
2. **Strip sensitive markers**: Remove features like names, gendered language, and affiliations that signal gender identity before training the model.
3. **Add fairness constraints**: Use fairness-aware algorithms or post-processing techniques to ensure balanced outcomes in resume scoring.

Q7: Suggest metrics to evaluate fairness after fixing.

- **Disparate Impact Ratio**: Measures whether different groups are receiving favorable outcomes at similar rates.
- **Equal Opportunity Difference**: Checks if qualified candidates from all groups have equal chances of being recommended.
- **False Negative Rate by group**: Ensures the system isn't missing strong candidates disproportionately based on gender.

Case 2: Facial Recognition in Policing

Scenario: A facial recognition system misidentifies minorities at higher rates.

Q8: What are the ethical risks?

There are several major concerns. First, the system can lead to **wrongful arrests**, especially when the technology misidentifies people of color more often. This can ruin lives and create fear and mistrust between communities and law enforcement. Second, there's the issue of **privacy violations**, since facial data can be collected and stored without people's consent. That creates a surveillance environment where people feel watched and targeted, even if they've done nothing wrong. Finally, there's a **lack of accountability**. If an AI makes a mistake, it's often hard to trace who's responsible or how the error happened.

Q9: What policies would help guide responsible use?

1. **Mandatory bias testing**: Regular audits should be conducted to measure error rates across racial and ethnic groups before deployment.
2. **Transparent usage protocols**: Agencies should publicly share how the technology is used, including which cases involve facial recognition.
3. **Informed consent and opt-outs**: People should know when their facial data is being collected and have the right to refuse unless it's legally required.
4. **Oversight committees**: Independent bodies should monitor how law enforcement uses these tools and ensure fairness in their outcomes.

PART 4

Ethical Reflection – AI Principles in Practice

In a previous project, I explored bias mitigation in the COMPAS dataset, which raised pressing questions around fairness, accountability, and transparency in criminal justice predictions. Going forward, I plan to extend this into a more holistic fairness assessment tool ,one that not only diagnoses bias but explains its roots and suggests actionable policy pathways.

To ensure this future project adheres to ethical AI principles, I'll ground it in the following commitments:

Fairness

- **Minimize discrimination** by carefully identifying protected attributes and validating fairness metrics like disparate impact and equal opportunity.
- Include multiple demographic slices to avoid overlooking smaller or intersectional groups.

Accountability

- Maintain **transparent documentation** for every model decision — what data was used, how it was processed, and what mitigation techniques were applied.
- Invite **peer feedback and audits**, especially from stakeholders who represent affected communities.

Transparency

- Avoid "black-box" behavior by using interpretable models or explainability tools like SHAP or LIME.
- Clearly communicate to users (or policymakers) what the model does, doesn't do, and how confident it is in its predictions.

Privacy

- Respect individual data rights by anonymizing input wherever possible and minimizing personally identifiable information.
- Stay current with data governance regulations, and seek consent or institutional review when analyzing sensitive datasets.

Purpose Alignment

- Ensure the project's goal aligns with socially beneficial outcomes — never automating decisions that should remain human-centered without proper safeguards.
- Regularly revisit the project's ethical posture, especially as its usage evolves.

BONUS TASK

Ethical AI Use in Healthcare: Policy Guideline

1. Patient Consent Protocols

- **Informed Consent**: All patients must be clearly informed when AI tools are involved in diagnosis, treatment recommendations, or data analysis. Consent should include explanation of the AI's role, limitations, and data usage.
- **Opt-in/Opt-out Choice**: Patients must be given the choice to opt in or out of AI-assisted care without penalty. Consent should be revocable at any point.
- **Data Protection Assurance**: Patient data used by AI systems must be de-identified and stored securely. Usage beyond direct care (e.g. for model training or research) requires explicit additional consent.
- **Human Oversight**: Consent must emphasize that AI decisions do not replace licensed medical judgment, humans remain ultimately responsible for care.

2. Bias Mitigation Strategies

- **Representative Training Data**: AI models should be trained on diverse datasets that reflect population demographics, including race, gender, age, and socioeconomic status, to avoid systemic bias.

- **Fairness Audits**: Regular evaluations of model outputs for disparities in treatment, access, or outcomes across different groups. Any bias detected should trigger retraining or mitigation.
- **Stakeholder Inclusion**: Healthcare workers, ethicists, and patient advocates must be involved in reviewing AI model design and its impact on vulnerable groups.
- **Feedback Mechanisms**: Provide a way for users to flag concerns or misbehavior in the AI system, with follow-up procedures in place.

3. Transparency Requirements

- **Explainability by Design**: AI systems used in healthcare must produce understandable reasoning for their predictions, whether via transparent logic or post hoc explanation tools like SHAP.
- **Clear Documentation**: Every AI model must have documented data sources, training objectives, known limitations, and change logs. This documentation must be accessible to oversight bodies.
- **Disclosure Standards**: Institutions must disclose when and where AI is deployed in patient care workflows. Staff and patients should never be unaware of AI usage.
- **Performance Benchmarks**: Models must undergo validation with published results before being used clinically. Benchmarks should include accuracy, fairness, and safety metrics.